Ethyl alcohol (No. Lab. Preⁿ)

1. Preparation → When Ethyl Iodide is treated with KOH aq we get C_2H_5OH.

$$C_2H_5I + \underset{(aq)}{KOH} \rightarrow KI + C_2H_5OH$$

W.H

2. When Ethyl acetate is boiled with KOH(aq) we get C_2H_5OH & CH_3COOK

$$CH_3COOC_2H_5 + KOH \rightarrow CH_3COOK + C_2H_5OH$$

3. When Acetaldehyde is reduced by hydrogen we get Ethanol.

$$\begin{matrix} CH_3 \\ H \end{matrix}\!\!>C=O + 2H \rightarrow \begin{matrix} CH_3 \\ H \end{matrix}\!\!>C<\!\!\begin{matrix} OH \\ H \end{matrix}$$

or (C_2H_5OH)

W.H

4. When Ethyl amine is treated with HNO_2 (Nitrous acid) / $NaNO_2 + HCl$ we get Ethanol.

$$C_2H_5NH_2 + OHNO \rightarrow C_2H_5OH + N_2 + H_2O$$

Industrial Preparation

It is prepared by the fermentation of Sugar.

$$C_{12}H_{22}O_{11} \xrightarrow[\text{(Invertage)}]{\text{Fermantation} + H_2O} 2C_6H_{12}O_6$$

(Glucose / Fructose)

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} 2CO_2 + 2C_2H_5OH$$

Properties

Physical → It is a colourless liquid Soluble in water, a good solvent for organic compounds. Good oudure. It causes excitement but not poiseneous and is called wine in dilute form.

CHEMICAL PROPE

1. Oxidation / Burning → ($C_2H_5OH + 3O_2$)

$$C_2H_5OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O$$

2. With ($K_2Cr_2O_7$ + H_2SO_4) → due to oxidation CH_3CHO is formed but on prolong oxidation CH_3COOH is resulted.

$$C_2H_5OH + O \xrightarrow[\text{(acidified } K_2Cr_2O_7)]{} CH_3CHO + H_2O$$

$$CH_3CHO + O \longrightarrow CH_3COOH$$

3. With Na –

$$C_2H_5OH + Na \longrightarrow C_2H_5ONa + \tfrac{1}{2}H_2$$

4. With HCl gas – in the presence of anhydrous $ZnCl_2$ it reacts with HCl to form C_2H_5Cl

$$C_2H_5OH + HCl \xrightarrow[\text{(Anhydrous } ZnCl_2)]{} C_2H_5Cl + H_2O$$

(5) With H_2SO_4 – (last page)

(6) With Acetic acid → It gives Ester / Ethyl acetate

$$CH_3COOH + HOC_2H_5 \longrightarrow CH_3COOC_2H_5 + H_2O$$

(7) Paste of Bleching Powder – (See Choloroform)

(8) With Iodine & NaOH → (See Iodoform)

(9) With PCl_5 – It gives Ethyl Choloride.

$$C_2H_5OH + PCl_5 \longrightarrow POCl_3 + HCl + C_2H_5Cl$$

(10) With Cholorine water → It is oxidised by Cholorine water.

$$C_2H_5OH + O \longrightarrow CH_3CHO + H_2O$$

$$CH_3CHO + 3Cl_2 \longrightarrow CCl_3CHO + 3HCl$$

Uses OF Ethyl alcohol.

1. Good Solvent
2. To prepare Choloroform, Iodoform etc.
3. To prepare perfumes, ~~transperent~~ Soap.
4. As a Methylated Spirit.

1. (i) Write notes on (1) Methylated Spirit (2) Absolute alcohol (3) Power Alcohol

(ii) In a tabular form distinguish CH_3OH and C_2H_5OH. Convert Methyl alcohol to Ethyl alcohol and vice versa.

(iii) Prove that C_2H_5OH, Contains C, and H_2.

Ans (1) Methylated spirit → It is a mixture of ethyl alcohol, Methyl alcohol, fusel oil, meneral naptha, piridine to denature the alcohols the above chemicals are added. Now a days a little copper Sulphate (poison) is added to Methylated Spirit So that nobody can drink it. It is used for varnishes liquid fuel and for Sprit lamp. It is expected that the percentage of Methyl alcohol is nearly 50%.

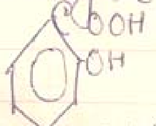
(2) Absolute alcohol → Rectified Spirit Contains nearly 95% alcohol and rest water. It may be converted into 99.7% alcohol this is known as Absolute alcohol.

When Rectified Spirit is distilled with Calcium oxide we get absolute alcohol in the receiver.

(3) Power alcohol → It is a mixture of Absolute alcohol, Petrol and Benzene this mixture gives more energy to the Arrowplane engine

then if only petol is used so it is a special type of liquid fuel for Airships because only petrol accupies more space in aircraft.

Distinction between Ethyl alcohol & Methyl alcohol.

Reagent	Methyl alcohol	Ethyl alcohol
1. Physiological action	It causes blindness if taken orally.	It causes exitement and giddiness.
2. With - Sodium	It gives Sodium Methoxide	It gives Sodium Ethoxide.
3. With PCl_5	Methylcholoride is formed	Ethyl Cholorid e is formed.
4. With I_2 and NaOH warmed at 60°C	No Iodoform is formed.	Iodoform is formed.
5. With Acetic acid and few drops of Conc. H_2SO_4 Catalyst.	Methyl acetate is formed	Ethyl acetate is formed.
6. With Salicylic acid  + a few drops Conc H_2SO_4, heating	Very nice cent of Methyl Salicylate or (winter green) is obtained.	No such nice oudure.

Convert Ethyl alcohol to Methyl alcohol and Vix versa.

(a) C_2H_5OH to CH_3OH (Steam clown)

When Ethyl alcohol is oxidised by $K_2Cr_2O_7$ and H_2SO_4 we get acetaldehyde. This on further oxidation with excess of the above reagent we get CH_3COOH. This is nutralyzed by NaOH Sol^n to get Sodium acetate Salt. On heating the Sodium acetate with Sodalime due to decarboxilation Methene is formed. Methane reacts with Cholorine in Sun light in a Controlled way to form CH_3Cl. On heating with Aquaus KOH we get Methanol.

$$C_2H_5OH \xrightarrow[(K_2Cr_2O_7 + H_2SO_4)\ -H_2O]{O} CH_3CHO \xrightarrow[(K_2Cr_2O_7 + H_2SO_4)]{+O} CH_3COOH$$

$$CH_3COOH \xrightarrow[-H_2O]{+NaOH} CH_3COONa$$

$$CH_3COONa \xrightarrow[-Na_2CO_3]{+NaOH} CH_4 \xrightarrow[(Controlled, S.L)\ -HCl]{Cl_2} CH_3Cl$$

$$CH_3Cl \xrightarrow[-KCl]{+KOH} CH_3OH$$

(b) CH_3OH to C_2H_5OH

When Methyl alcohol is treated with PCl_5 we get Methyl cholorde. This Methyl Cholorde is heated with metalic Sodium in the presence of Ether, (Wurtz reaction) we get Ethene. Ethane reacts with Cholorine in a controlled way in Sun light to get C_2H_5Cl.

On treating with KOH Aq. we get Ethanol.

$$CH_3OH \xrightarrow[-POCl_3,\ -HCl]{+PCl_5} CH_3Cl \times 2 \xrightarrow[-2NaCl]{+2Na} C_2H_6$$

$$C_2H_6 \xrightarrow[-HCl]{+Cl_2 \text{ Controlled}} C_2H_5Cl$$

$$C_2H_5OH \xleftarrow[-KCl]{KOH_{(aq)}} C_2H_5Cl$$